

Certificate PDF — texture recompression

Before / after comparison for your approval — nothing has been changed yet

The file public/certificates/ysf-internship-certificate.pdf is 4.54 MB. 92.6% of it (4.20 MB) is a single paper-texture JPEG. Everything you can read — CERTIFICATE, the name, the dates, the signature — is vector text drawn on top of that texture, so re-compressing the texture cannot blur any text.

Two facts made this cheap to fix: the texture is stored at 4608x3376 but drawn 1.96x wider than the page, so 47% of those pixels were never visible; and it is painted at about 79 DPI on paper, so it only needs screen-level detail.

Measured results

option	texture pixels	file size	vs before	render match vs original
BEFORE (current file)	4608 x 3376 (4.20 MB JPEG)	4,539,745 B	—	reference
C1 crop only, q95	2463 x 3376	2,331,265 B	-49%	47-51 dB PSNR, max diff 22/255
C2 crop + 0.70x, q88	1724 x 2363	982,796 B	-78%	47.1 dB PSNR, max diff 21/255
C4 crop + 0.40x, q85	985 x 1350	511,521 B	-89%	42.9 dB PSNR, max diff 27/255

PSNR above 45 dB is generally considered visually indistinguishable for photographic content; the "max diff" column is the worst single pixel in an 8.7-million-pixel page. C2 changed 0.010% of pixels by more than 8/255 at 200 DPI; C4 changed 0.114%.

Recommendation

C2 (crop + 0.70x, q88) is the pick: the certificate drops from 4.54 MB to 0.98 MB (-78%), the paper texture stays natural at 1:1, and the zoomed comparison on page 2 shows no visible difference. C4 saves 0.47 MB more but does start to flatten the grain if you look closely — fine for a web preview, slightly risky for a document you may want to print.

What changes, what stays exactly the same

Stays identical

- Every word and letter — all text is vector, untouched
- Fonts, layout and page size (A4, 596 x 850 pt)
- The signature image (499x500 px), unchanged
- Tagged-PDF structure and accessibility tree
- The certificate stays a single-page PDF

Changes

- The paper texture only, re-encoded as JPEG
- Its stored pixel size (downscaled in C2/C4)
- Nothing else in the file is touched

How each candidate was verified

- Rendered the original and the candidate through the same PDF engine at 150, 200 and 300 DPI.
- Compared every pixel: max difference, mean difference, share of pixels off by more than 8/255, and PSNR.
- Checked the vector layers separately by zooming to 300 DPI on the heading block (page 4): the glyphs are identical.
- Confirmed the file still parses as a tagged PDF with no syntax errors (pikepdf/qpdf check_pdf_syntax).

If you say go

I apply the chosen option to `public/certificates/ysf-internship-certificate.pdf`, re-run `npm run check:meta` (metadata stays clean), rebuild, and report the new size. Nothing else in S6 depends on this, and I will not touch the file until you pick — C2, C4, or leave it alone.

BEFORE — 4,519,745 B (4.54 MB) IC/25/2669 25.11.2025  CERTIFICATE OF COMPLETION This is to certify that ASHISH KUMAR HAS COMPLETED HIS TENURE AS A SOCIAL MEDIA MARKETING INTERN AT YATHAARTH SANGAM FOUNDATION FROM SEPTEMBER 2025 TO NOVEMBER 2025 WE APPRECIATE HIS VALUABLE CONTRIBUTIONS AND WISH HIM ALL THE BEST IN HIS FUTURE ENDEAVORS. Sujay Kumar NAME CO-FOUNDER Harsh Raj NAME Founder	AFTER C2 — 982,796 B (0.98 MB, -78%) IC/25/2669 25.11.2025  CERTIFICATE OF COMPLETION This is to certify that ASHISH KUMAR HAS COMPLETED HIS TENURE AS A SOCIAL MEDIA MARKETING INTERN AT YATHAARTH SANGAM FOUNDATION FROM SEPTEMBER 2025 TO NOVEMBER 2025 WE APPRECIATE HIS VALUABLE CONTRIBUTIONS AND WISH HIM ALL THE BEST IN HIS FUTURE ENDEAVORS. Sujay Kumar NAME CO-FOUNDER Harsh Raj NAME Founder	AFTER C4 — 511,521 B (0.51 MB, -89%) IC/25/2669 25.11.2025  CERTIFICATE OF COMPLETION This is to certify that ASHISH KUMAR HAS COMPLETED HIS TENURE AS A SOCIAL MEDIA MARKETING INTERN AT YATHAARTH SANGAM FOUNDATION FROM SEPTEMBER 2025 TO NOVEMBER 2025 WE APPRECIATE HIS VALUABLE CONTRIBUTIONS AND WISH HIM ALL THE BEST IN HIS FUTURE ENDEAVORS. Sujay Kumar NAME CO-FOUNDER Harsh Raj NAME Founder
--	--	--

BEFORE — 300 DPI, 1:1

CERTIFICATE

OF COMPLETION

This is to certify that

AFTER C2 (0.70x q88) — same crop

CERTIFICATE

OF COMPLETION

This is to certify that

AFTER C4 (0.40x q85) — same crop

CERTIFICATE

OF COMPLETION

This is to certify that

BEFORE — 200 DPI, 1:1 texture patch



AFTER C2 (0.70x q88) — same patch



AFTER C4 (0.40x q85) — same patch



Page 5 — paper texture at 1:1

Left, top to bottom: the original texture, C2, C4 — the same patch of paper at 200 DPI, shown at actual size. C2 keeps the grain; C4 smooths it slightly, which is what the extra 0.47 MB buys.